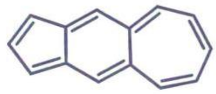
2025-2026 合成化学 I 期中测试

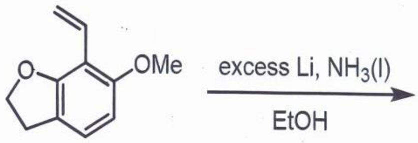
2025.11.8　　14:30-16:30

整理：Aureliano_Drogo

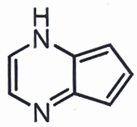
1. Explain why the following molecule has a large dipole moment（偶极矩）. (4 pts)



2. Predict the product for the following **Birch reduction** and draw the mechanism. (4 pts)

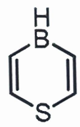


3. Are the following compounds **aromatic**, **antiaromatic**, or **nonaromatic**? Justify your answer. (4 pts)

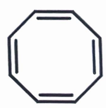


A

B

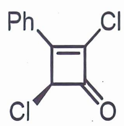


C

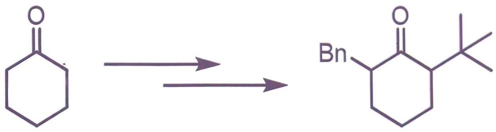


D

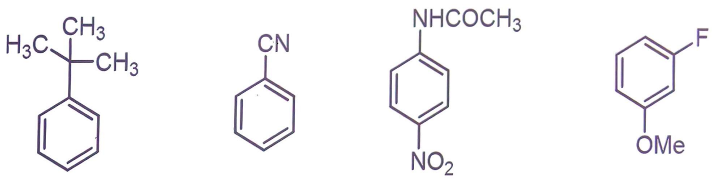
4. The optically active dichlorophenylcyclobutenone A undergoes racemization (消旋化) in acetic acid at 100 °C. Please provide an explanation. (4 pts)



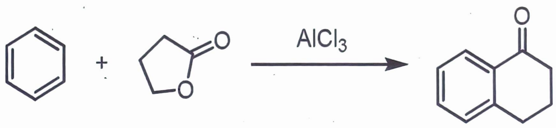
5. Using necessary organic and inorganic agents, complete the following transformation. (4 pts)



6. Indicate the major position(s) of **catalyzed bromination** for the following compounds. (4 pts, 1 pt for each)

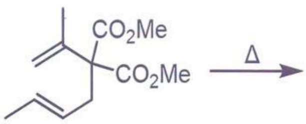


7. Propose a mechanism for the following reaction. (4 pts)

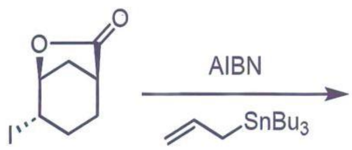


8. Draw the major products for the following reactions. (6 pts, 2 pts for each)

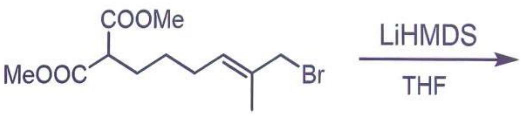
1)



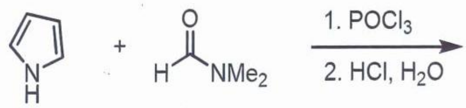
2)

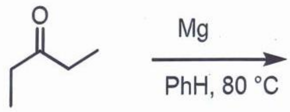


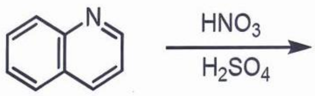
3)



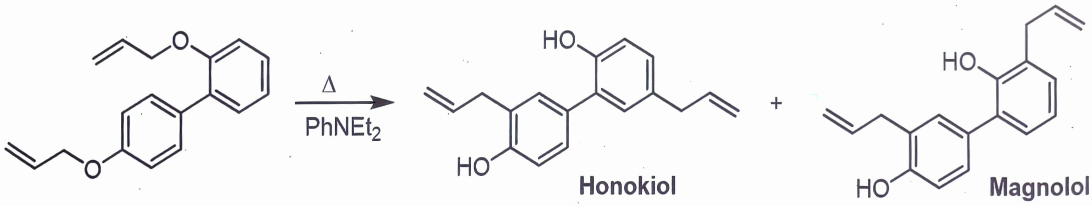
9. Draw the major products for the following reactions. (6 pts, 2 pts for each)

1) 

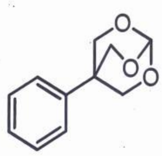
2) 

3) 

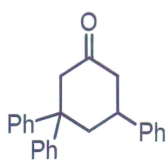
10. Suggest a mechanism for the following reaction. (5 pts)



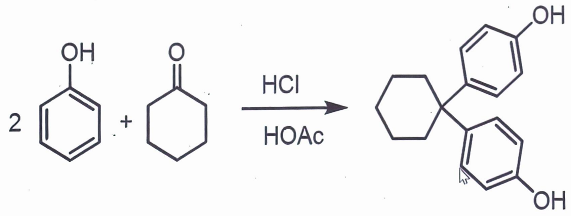
11. Suggest a synthetic plan for the following compound with **benzene**, **toluene**, and/or organic compounds containing **four or fewer carbon atoms** and necessary inorganic reagents. (5 pts)



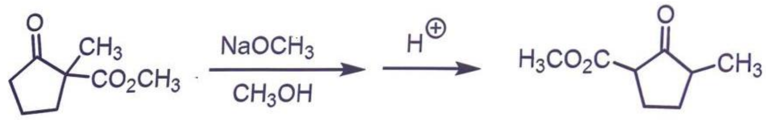
12. Suggest a synthetic plan for the following compound with **benzene** and organic starting materials containing **four or fewer carbon atoms** and necessary inorganic reagents. (5 pts)



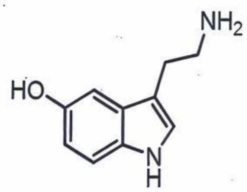
13. Propose a mechanism for the following reaction. (5 pts)



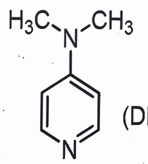
14. *Suggest a mechanism for the following reaction. (5 pts)*



15. Propose a synthetic approach for serotonin, which is a common neurotransmitter, using **phenol**, organic reagents containing **four or fewer carbon atoms**, and necessary inorganic reagents. (5 pts)

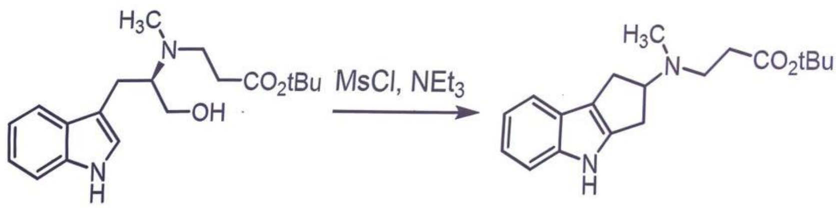


16. Propose a synthetic approach from **pyridine** for DMAP. (5 pts)



(DMAP)

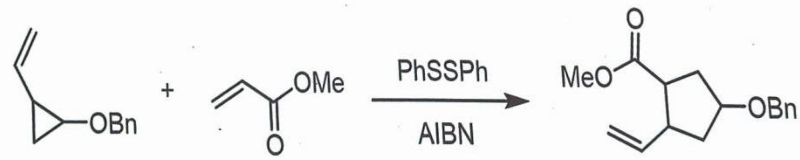
17. Suggest a mechanism for the following reaction. Note the stereochemistry. (5 pts)



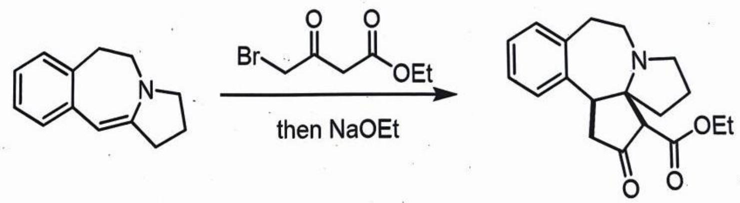
Optically pure

Racemic

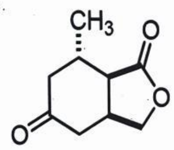
18. Suggest a mechanism for the following reaction. (5 pts)



19. Suggest a mechanism for the following reaction. (5 pts)



20. Suggest a synthetic plan for the following compound with organic starting materials containing **four or fewer carbon atoms** and necessary inorganic reagents. (5 pts)



21.Propose a mechanism for the following pericyclic reaction and draw the HOMO of the active intermediate to explain the stereoselectivity of the reaction.(5 pts)